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His door is always open

إِلَـهِي غَارَتِ النُّجُوْمُ، وَنَامَتِ العُيُوْنُ وَغَلَّقَتِ المُلُوْكُ
أَبْوَابَهَا، وَبَابُكَ مَفْتُوْحٌ، وَخَلَا كُلُّ حَبِيْبٍ بِحَبِيْبِهِ،
وَهَذَا مَقَامِي بَيْنَ يَدَيْكَ

O Allah, the stars have vanished, the eyes have slept, the kings have locked their doors, and Your door remains open. Every lover has found privacy with their beloved, and here I am standing before You.

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On-premise vs. Cloud



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Reference https://www.cleo.com/blog/knowledge-base-on-premise-vs-cloud a good article to read.

Cloud security will be covered in a lot of detail in the advanced CODS to be released in 2021 inshAllah.

In today's world of enterprise IT, there are many factors that a company must consider in order to decide whether a cloud infrastructure is the right fit. Conversely, there are many companies that are unable make the leap into the cloud, instead relying on their tried-and-true legacy and on-premise applications and software to do business.

On Premise vs. Cloud

It's no surprise that cloud computing has grown in popularity as much as it has, as its allure and promise offer newfound flexibility for enterprises, everything from saving time and money to improving agility and scalability. On the other hand, on-premise software – installed on a company's own servers and behind its firewall – was the only offering for organizations for a long time and may continue to adequately serve your business needs (think, "if it ain't broke then don't fix it"). Additionally, on-premise applications are reliable, secure, and allow enterprises to maintain a level of control that the cloud often cannot. But there's agreement among IT decision-makers that in addition to their on-premise and legacy systems, they'll need to leverage new cloud and SaaS applications to achieve their

business goals.

On-Premise Software

Whether a company places its applications in the cloud or whether it decides to keep them on premises, data security will always be paramount. But for those businesses in highly regulated industries, the decision might already be made for them as to whether to house their applications on premise. And knowing your data is located within your in-house servers and IT infrastructure might also provide more peace of mind anyway.

On-premise software requires that an enterprise purchases a license or a copy of the software to use it. Because the software itself is licensed and the entire instance of software resides within an organization's premises, there is generally greater protection than with a cloud computing infrastructure. So, if a company needs all this extra security, why would they dip its proverbial toes into the cloud?

The downside of on-premise environments is that costs associated with managing and maintaining all the solution entails can run exponentially higher than a cloud computing environment. An on-premise setup requires in-house server hardware, software licenses, integration capabilities, and IT employees on hand to support and manage potential issues that may arise. This doesn't even factor in the amount of maintenance that a company is responsible for when something breaks or doesn't work.

Cloud Computing

Cloud computing differs from on-premises software in one critical way. A company hosts everything in-house in an on-premise environment, while in a cloud environment, a third-party provider hosts all that for you. This allows companies to pay on an as-needed basis and effectively scale up or down depending on overall usage, user requirements, and the growth of a company.

A cloud-based server utilizes virtual technology to host a company's applications offsite. There are no capital expenses, data can be backed up regularly, and companies only have to pay for the resources they use. For those organizations that plan aggressive expansion on a global basis, the cloud has even greater appeal because it allows you to connect with customers, partners, and other businesses anywhere with minimal effort.

Additionally, cloud computing features nearly instant provisioning because everything is already configured. Thus, any new software that is integrated into your environment is ready to use immediately once a company has subscribed. With instant provisioning, any time spent on installation and configuration is eliminated and users are able to access the application right away.

Key Differences of On Premise vs. Cloud

As outlined above, there are a number of fundamental differences between an on-premises and a cloud environment. Which path is the correct one for your enterprise depends entirely on your needs and what it is you're looking for in a solution.

Deployment

On Premises: In an on-premises environment, resources are deployed in-house and within an enterprise's IT infrastructure. An enterprise is responsible for maintaining the solution and all its related processes.

Cloud: While there are different forms of cloud computing (such as public cloud, private cloud, and a hybrid cloud), in a public cloud computing environment, resources are hosted

on the premises of the service provider but enterprises are able to access those resources and use as much as they want at any given time.

Cost

On Premises: For enterprises that deploy software on premise, they are responsible for the ongoing costs of the server hardware, power consumption, and space.

Cloud: Enterprises that elect to use a cloud computing model only need to pay for the resources that they use, with none of the maintenance and upkeep costs, and the price adjusts up or down depending on how much is consumed.

Control

On Premises: In an on-premises environment, enterprises retain all their data and are fully in control of what happens to it, for better or worse. Companies in highly regulated industries with extra privacy concerns are more likely to hesitate to leap into the cloud before others because of this reason.

Cloud: In a cloud computing environment, the question of ownership of data is one that many companies – and vendors for that matter, have struggled with. Data and encryption keys reside within your third-party provider, so if the unexpected happens and there is downtime, you maybe be unable to access that data.

Security

On Premises: Companies that have extra sensitive information, such as government and banking industries must have a certain level of security and privacy that an on-premises environment provides. Despite the promise of the cloud, security is the primary concern for many industries, so an on-premises environment, despite some if its drawbacks and price tag, make more sense.

Cloud: Security concerns remain the number one barrier to a cloud computing deployment. There have been many publicized cloud breaches, and IT departments around the world are concerned. From personal information of employees such as login credentials to a loss of intellectual property, the security threats are real.

Compliance

On Premises: Many companies these days operate under some form of regulatory control, regardless of the industry. Perhaps the most common one is the Health Insurance Portability and Accountability Act (HIPAA) for private health information, but there are many others, including the Family Educational Rights and Privacy Act (FERPA), which contains detailed student records, and other government and industry regulations. For companies that are subject to such regulations, it is imperative that they remain compliant and know where their data is at all times.

Cloud: Enterprises that do choose a cloud computing model must do their due diligence and ensure that their third-party provider is up to code and in fact compliant with all of the different regulatory mandates within their industry. Sensitive data must be secured, and customers, partners, and employees must have their privacy ensured.

Hybrid Cloud Solutions

While the debate of the pros and cons of an on-premises environment pitted against a cloud computing environment is a real one, and one that many enterprises are having within their offices right now, there is another model that offers the best of both worlds.

A hybrid cloud solution is a solution that features an element of different types of IT

deployment models, ranging from on premises to private cloud and public cloud. A hybrid cloud infrastructure depends on the availability of a public cloud platform from a trusted third-party provider, a private cloud constructed either on premises or through a hosted private cloud provider, and effective WAN connectivity between both of those environments.

Cleo Integration Cloud

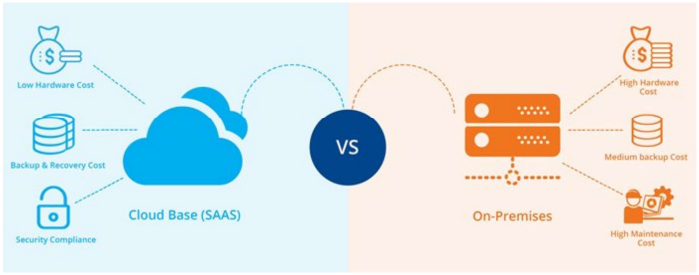
Regardless of what kind of environment you are looking for, whether that's to add a software-as-a-service (SaaS) solution to address a specific business need, move processes and data into a cloud integration platform, or whether you are a SaaS organization that thrives on delivering faster responses to customer requests, you rely on integration to make your data flows work.

Every successful company needs a scalable infrastructure that can support any-to-any hybrid integration, data transformation, fast and secure file transfer, and end-to-end visibility of all the data that flows their dynamic ecosystems. Cleo Integration Cloud enables enterprises to accelerate ground-to-cloud and cloud-to-cloud integration processes to easily integrate applications, and storage and business platforms, to connect all your data, no matter what it is, and wherever you want it, be it on premises or in the cloud.

Cleo Integration Cloud also features self-service and managed services for business and technical users alike, and allows them to build, control, and monitor any and all B2B, application, cloud integration, and data lake ingestion processes.

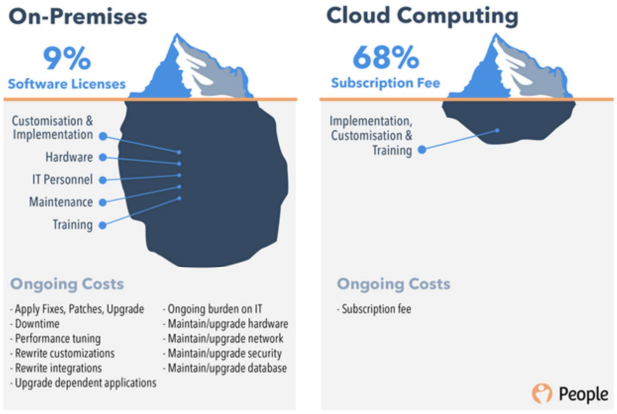
Contact Cleo today to learn more about integrating the critical on-premise and cloud applications running your business.

On-Premise vs Cloud Based what is it?



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So what is On-Premise Computing vs. Cloud Computing



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Definition of Cloud Computing

For the sake of discussion, cloud computing has been formally defined by NIST as:

"… a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

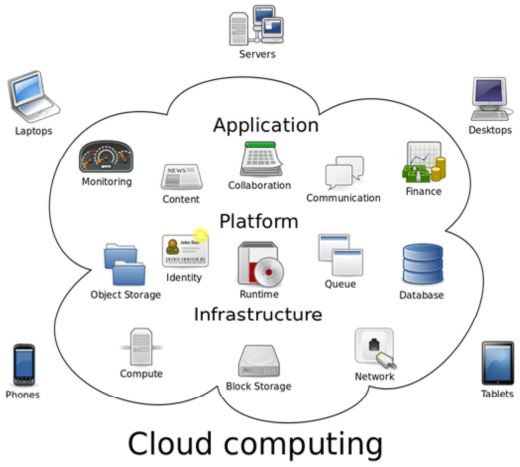
The definition from the comparable ISO/IEC standard 17888 for cloud computing is similar:

"Paradigm for enabling network access to a scalable and elastic pool of shareable physical or virtual resources with self-service provisioning and administration on-demand."

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Cloud-based Systems

- On-Demand Self Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service
- Multi-Tenancy



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NIST defines the five essential characteristics of cloud computing as the following:

On-Demand Self-Service: A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

Broad Network Access: Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

Resource Pooling: The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. Examples of resources include storage, processing, memory, and network bandwidth.

Rapid Elasticity: Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.

Measured Service: Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).

Multi-Tenancy: A feature where physical or virtual resources are allocated in such a way that multiple tenants and their computations and data are isolated from and inaccessible to one another.

Cloud Based System Models

- Software as a service (SaaS)
- Platform as a service (PaaS)
- Infrastructure as a service (IaaS)
- Network as a service (NaaS)

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Reference https://www.fingent.com/blog/cloud-service-models-saas-iaas-paas-choose-the-right-one-for-your-business

The future of computing is in the cloud. What it implies is that you adapt your business to fit in the cloud model. The opposite also holds true as your business can be left behind if this new technology is underutilized.

Once you sign up for cloud service models Saas, IaaS, Paas , you can leverage its wider possibilities to bring the flexibility and efficiency that pushes your business growth.

Over the years cloud services have witnessed an exponential growth worldwide. Gartner's forecast of global public cloud services revenue estimates that 2018 alone will generate 305.8 billion dollars with 411.4 billion dollars growth projected at 2020.

What these reports signify is a steady adoption of cloud services by businesses across the world to tackle the entire range of operations that they do. Meanwhile, numerous leading players in the information technology sector now compete to deliver flexible cloud services for both the public and enterprises.
Increasing competition meant better delivery of services and innovations, which could deeply benefit you in scaling up your business. Hence, now is the right time that you deploy

a cloud model into your business infrastructure.

An Overview of Cloud Benefits

By adopting a cloud service into your enterprise, what could it possibly do to widen the scope of your operations? The pros of cloud adoption far outweigh its cons, which is one reason why you should consider it in the first place. Some of its advantages include:

Scalable – A cloud service allows quick scaling up and down of computing resources to accommodate your changing needs.

Affordable – You pay less for a cloud service, as it eliminates unnecessary costs involved in hardware upgrades and maintenance.

Secure – By signing up for a cloud service, you are essentially making your data more secure using their industry-grade security protocols.

If you have envisioned a goal of making your business more dynamic, then the cloud is the way. And the question comes down to this: what type of cloud service model would you implement and which one will fit to your unique business requirements?

Cloud Service Models Saas, IaaS, PaaS

Cloud models come in three types: SaaS (Software as a Service), IaaS (Infrastructure as a Service) and PaaS (Platform as a Service). Each of the cloud models has their own set of benefits that could serve the needs of various businesses.

Choosing between them requires an understanding of these cloud models, evaluating your requirements and finding out how the chosen model can deliver your intended set of workflows. The following is a brief description of the three types of cloud models and their benefits.

SaaS

SaaS or Software as a Service is a model that gives quick access to cloud-based web applications. The vendor controls the entire computing stack, which you can access using a web browser. These applications run on the cloud and you can use them by a paid licensed subscription or for free with limited access.

SaaS does not require any installations or downloads in your existing computing infrastructure. This eliminates the need for installing applications on each of your computers with the maintenance and support taken over by the vendor. Some known example of SaaS includes Google G Suite, Microsoft Office 365, Dropbox etc.

IaaS

IaaS or Infrastructure as a Service is basically a virtual provision of computing resources over the cloud. An IaaS cloud provider can give you the entire range of computing infrastructures such as storage, servers, networking hardware alongside maintenance and support. Businesses can opt for computing resources of their requirement without the need to install hardware on their premises. Amazon Web Services, Microsoft Azure, and Google Compute Engine are some of the leading IaaS cloud service providers.

PaaS

Platform as a Service or PaaS is essentially a cloud base where you can develop, test and organize the different applications for your business. Implementing PaaS simplifies the process of enterprise software development. The virtual runtime environment provided by PaaS gives a favorable space for developing and testing applications.

The entire resources offered in the form of servers, storage and networking are manageable

either by the company or a platform provider. Google App Engine and AWS Elastic Beanstalk are two typical examples of PaaS. PaaS is also subscription based that gives you flexible pricing options depending on your business requirements.

Reference https://www.sdxcentral.com/networking/virtualization/definitions/what-is-naas/

Networking-as-a-service (NaaS) is the sale of network services from third parties to customers that don't want to build their own networking infrastructure.
NaaS packages networking resources, services, and applications as a product that can be purchased for a number of users, usually for a contracted period of time. It can include services such as Wide Area Networking (WAN) connectivity, data center connectivity, bandwidth on demand (Bandwidth On Demand), security services, and other applications.
Many forms of NaaS
Virtualization technology provides a platform for NaaS, which is related to other cloud services. Services are offered by Cloud Service Providers (Cloud Service Providers) in addition to NaaS include Software-as-a-service (SaaS), a computing platform for developing or hosting applications, known as Platform-as-a-service (PaaS) (PaaS); or an entire networking or computing infrastructure, known as Infrastructure-as-a-service (IaaS). Cloud Services such as NaaS and Paas are provided by building a large, scalable infrastructure that can be virtualized so that it can be sold to individual customers.
Source: NTTLarge NaaS providers includes major CSPs, including Amazon and Rackspace, as well as the global service providers such as AT&T, Level 3 Communications, Telefonica, and Verizon. More recently, niche NaaS providers have emerged in areas such as Software-Defined WAN, which includes players such as Aryaka, Cloudgenix, Pertino, and VeloCloud. It also includes specialized network providers such as Akamai, which has its own Content Delivery Network (CDN) for digital media delivery, as well as enterprise SaaS acceleration and security services..
Overall, NaaS applies to a broad set of applications and services. For example, Aryaka and Pertino offer WAN and secure Virtual Private Networks (VPN) as a service, Akamai offers CDN as a service, Amazon offers web-hosting, private cloud, and storage as a service, and many service providers offer Bandwidth On Demand and hosted networks as a service. Even entire service providers might outsource their networks, as in the case of mobile virtual network operator (MVNO).
Standards and SDN to Drive NaaS Growth
One of the challenges of NaaS is developing standards for network interoperability and portability. For example, a customer buying a NaaS service may want to make sure that if the service can be double-sourced or swapped out. In this case, the technology would need to be compatible with other platforms or standards. This is where important standards, APIs, and open-source initiatives such as OpenStack come into play. Existing Internet standards developed in the IEEE, such as MPLS and IP are important. The MEF and TM Forum are two standards working on interoperability between carrier NaaS offerings.
SDN is likely to make Networking- as-a-Service services more prevalent, as service providers look to leverage their hardware infrastructure so that it can be sold as enterprise network services. Virtualization enables the capability to share, market, or sell just about any

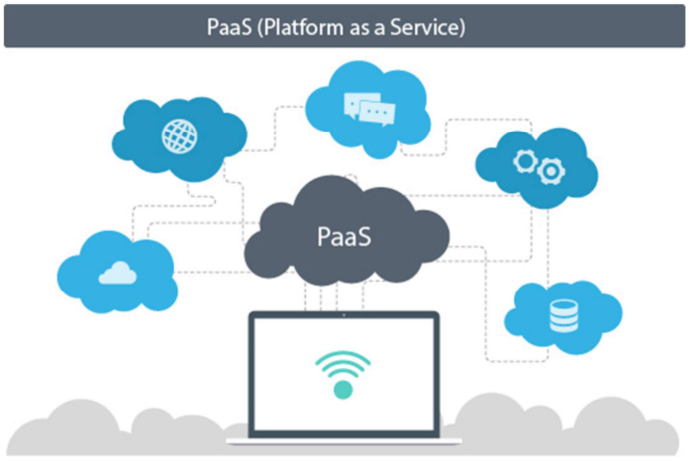
platform, whether it's cloud infrastructure, networking, or business applications.

Software as a service (SaaS)



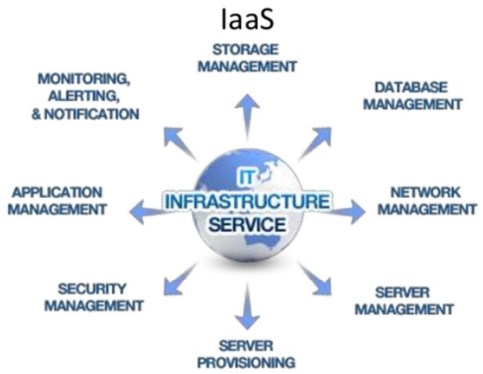
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Platform as a service (PaaS)



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Infrastructure as a service (IaaS)

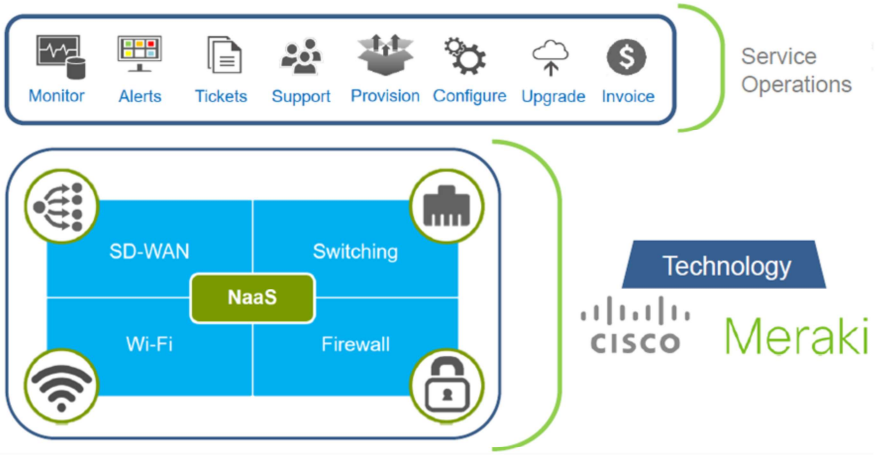


By Prof. Raj Sarode 4

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Network as a service (NaaS)



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